Case Report

THE NOVEL USE OF STEREOTACTIC RADIOTHERAPY FOR REMNANT ADRENAL TISSUE IN NELSON SYNDROME

Alexandra N. Martirossian, MD[1]; Rojymon Jacob, MD[2]; Kristen O. Riley, MD[3]; John R. Porterfield, MD[4]; Tom B. Vaughan, MD[5]

ABSTRACT

Objective: Recurrent Cushing disease (CD) is a rare complication that occurs in patients who have undergone bilateral adrenalectomy (BLA). We report a case of recurrent CD in a patient with Nelson syndrome and adrenalectomy due to remnant adrenal tissue, and a novel treatment strategy using stereotactic body radiation therapy (SBRT) to the adrenal glands.

Methods: We report a case of recurrent CD in a woman with Nelson syndrome and adrenalectomy and describe her clinical course and management. We also include a literature review of CD management and adrenal radiation.

Results: The patient had persistent pituitary CD despite pituitary surgery and radiosurgery and underwent BLA. She developed recurrent CD due to a remnant adrenal gland post adrenalectomy. She then underwent SBRT to both adrenal beds to treat the remnant adrenal tissue. Her serum cortisol dropped rapidly after adrenal radiation and she experienced minimal side effects. She has been in remission for over 2 years.

Conclusion: This is the first reported case of recurrent CD in a patient post adrenalectomy that was successfully treated with SBRT to the remnant adrenal tissue. **(AACE Clinical Case Rep. 2020;6:e33-e36)**

Abbreviations:
ACTH = adrenocorticotropic hormone; **BLA** = bilateral adrenalectomy; **CD** = Cushing disease; **RT** = radiotherapy; **SBRT** = stereotactic body radiotherapy; **TSS** = transsphenoidal surgery

INTRODUCTION

Cushing disease (CD), a disorder of elevated cortisol secondary to an adrenocorticotropic hormone (ACTH)-secreting pituitary tumor, is typically first treated with transsphenoidal surgery (TSS). However, recurrences of CD can occur after TSS, and other treatment modalities such as radiotherapy and bilateral adrenalectomy (BLA) must often be utilized. Nelson syndrome is a complication that can occur in CD patients who undergo BLA, and is caused by progression of the corticotroph pituitary tumor. Recurrence of CD after BLA is extremely rare, and can be caused by the stimulation of remnant or ectopic adrenal tissue. Herein we present a case of recurrent CD in a patient with Nelson syndrome and BLA and a novel treatment strategy.

CASE REPORT

A 24-year-old Caucasian female was diagnosed with CD and treated with TSS. Postoperatively, she had panhypopituitarism with diabetes insipidus and was placed on full replacement therapy. Six years after surgery, she developed fatigue, hair loss, hirsutism, excessive sweating, and weight gain, similar to her original presentation. After

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From [1]Department of Internal Medicine, University of Alabama at Birmingham, [2]Department of Radiation Oncology, University of Alabama at Birmingham, [3]Department of Neurosurgery, University of Alabama at Birmingham, [4]Department of Surgery, University of Alabama at Birmingham, and [5]Division of Endocrinology, Diabetes, and Metabolism. University of Alabama at Birmingham, Birmingham, Alabama.
Address correspondence to Dr. Alexandra N. Martirossian, Department of Internal Medicine, University of Alabama at Birmingham, 1720 Second Avenue South, BDB 327, Birmingham, AL 35294-0012.
E-mail: alexandramartirossian@yahoo.com.
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carefully tapering her glucocorticoid replacement, she was found to have an elevated serum cortisol level after a 1 mg dexamethasone suppression test of 12.4 mcg/dL (normal, <2.0 mcg/dL), and several elevated urinary free cortisol (UFC) studies ranging from 58 to 104.8 mcg/24 hours (normal, 4.0 to 50.0 mcg/24 hours). Her plasma ACTH was 32 pg/mL (normal, 6 to 50 pg/mL). A brain magnetic resonance imaging scan (MRI) showed no evidence of recurrent pituitary tumor. Control with ketoconazole was attempted, but she was unable to tolerate dose escalation and this was discontinued. She underwent a second TSS with particular attention to the site of her original surgery. Pathology showed no evidence of residual or recurrent adenoma. She was restarted on hydrocortisone postoperatively which was discontinued after 2 months based on clinical stigmata of ongoing CD. Five months after her second TSS, she had an elevated 24-hour UFC of 105.6 mcg/24 hours and ACTH of 25 pg/mL. After discussing various treatment options including medicine and pituitary radiosurgery, the patient elected to undergo laparoscopic BLA 1 year after her second TSS. Adrenal pathology showed bilateral nodular adrenal hyperplasia and her cortisol levels became undetectable. Steroid replacement therapy with hydrocortisone and fludrocortisone was initiated.

Three years after BLA, she again developed CD symptoms including weight gain, acne, hypertension, moon face, buffalo hump, and supraclavicular fat pad enlargement and her steroids were gradually stopped. An evening salivary cortisol was elevated at 1.48 mcg/dL (normal, ≤0.09 mcg/dL) and plasma ACTH was 386 pg/mL (normal, 0 to 45 pg/mL). A computed tomography scan of the abdomen with contrast showed an absent right adrenal gland with stranding in the right adrenal bed suggestive of adrenal remnant or recurrence. In the left adrenal fossa, there was a 9 mm soft tissue density nodule suggestive of residual or ectopic adrenal tissue. An MRI brain with and without contrast showed a new, enlarged focus of enhancing tissue in the right cavernous sinus measuring 7 mm × 6 mm × 8 mm consistent with Nelson syndrome. The patient was diagnosed with CD recurrence secondary to stimulation of remnant adrenal tissue in Nelson syndrome. She underwent a third TSS in which pathology showed corticotroph adenoma but was not expected to be curative. Surgical removal of her remnant adrenal tissue was considered, but ultimately deferred due to concern for surgical risks related to the small volume of tissue, its proximity to important structures, and prior surgery at those sites. To prevent a recurrence of Nelson syndrome and to avoid any chance of recurrent CD, a noninvasive approach involving radiotherapy to both the pituitary and adrenal tissue was pursued. Four months after her third TSS, she underwent pituitary radiosurgery (20 Gy single fraction). Clinically, there was no suggestion of any improvement of her CD in the subsequent 2 months. At that point, she was treated with stereotactic body radiotherapy (SBRT) to bilateral adrenal remnants (Fig. 1). A dose of 39 Gy was delivered in 3 treatments on a linear accelerator. This regimen was chosen because it was found to be safe and convenient to the patient based on institution experience treating adrenal metastases. It was estimated that a total dose of 39 Gy would deliver a sufficient biologically equivalent dose for control of benign tumors. Within 24 hours after the second fraction of adrenal SBRT, her serum morning cortisol dropped from 107.4 mcg/dL (normal, 5.0 to 23.0 mcg/dL) before treatment to 1.6 mcg/dL (Fig. 2). The patient concurrently experienced fatigue and nausea typical of adrenal insufficiency. Hydrocortisone and fludrocortisone were resumed immediately and her symptoms resolved. She has been in remission for over 2 years.

DISCUSSION

Cushing syndrome (CS) is a condition characterized by excess systemic cortisol, and has multisystemic manifestations that include weight gain, central obesity, hypertension, diabetes mellitus, osteoporosis, and mood changes (1). A corticotrope pituitary adenoma is the most common etiology of ACTH-dependent CS, and this condition is specifically defined as Cushing's disease (CD). The first-line treatment of CD is pituitary tumor resection via TSS (2). Remission rates of CD have been extensively studied and range from 69 to 93%, and tend to be better for microadenomas than macroadenomas (73 to 76% versus 43%) (2,3). Recurrence rates in adults are high and can vary from 15 to 66% within 5 to 10 years of initially successful surgery with a median or mean time to recurrence ranging from 20 to 84 months (2,3). Because of this risk for recurrence, regular surveillance is essential.

Once a diagnosis of recurrence has been made, one of several second line therapies can be utilized. In patients with evidence of incomplete tumor resection or a pituitary lesion on imaging, repeat TSS is recommended. Repeat TSS tends to have a lower remission rate than primary surgery, and carries with it a higher risk of panhypopituitarism (2,3). Medical therapy for CD may consist of steroidogenesis inhibitors such as ketoconazole, metyropone, mitotane, and etomidate, or pituitary-directed treatments such as pasireotide or cabergoline (1-3). Radiotherapy (RT) to the pituitary is another second line therapy for CD, commonly delivered using photon beams over several fractions (2). Stereotactic RT is the technique of delivering highly focused radiation precisely to the target, typically in less than 5 fractions. By using multiple ports to deliver radiation to the target, the dose to surrounding structures is minimized. When delivered in a single fraction, the technique is referred to as "radiosurgery." Comparing the outcomes of these radiation modalities for CD has been difficult due to variations in methods, definitions of biochemical and tumor control, and lengths of follow-up, so it is currently unclear which modality is superior. In addition, adjunctive

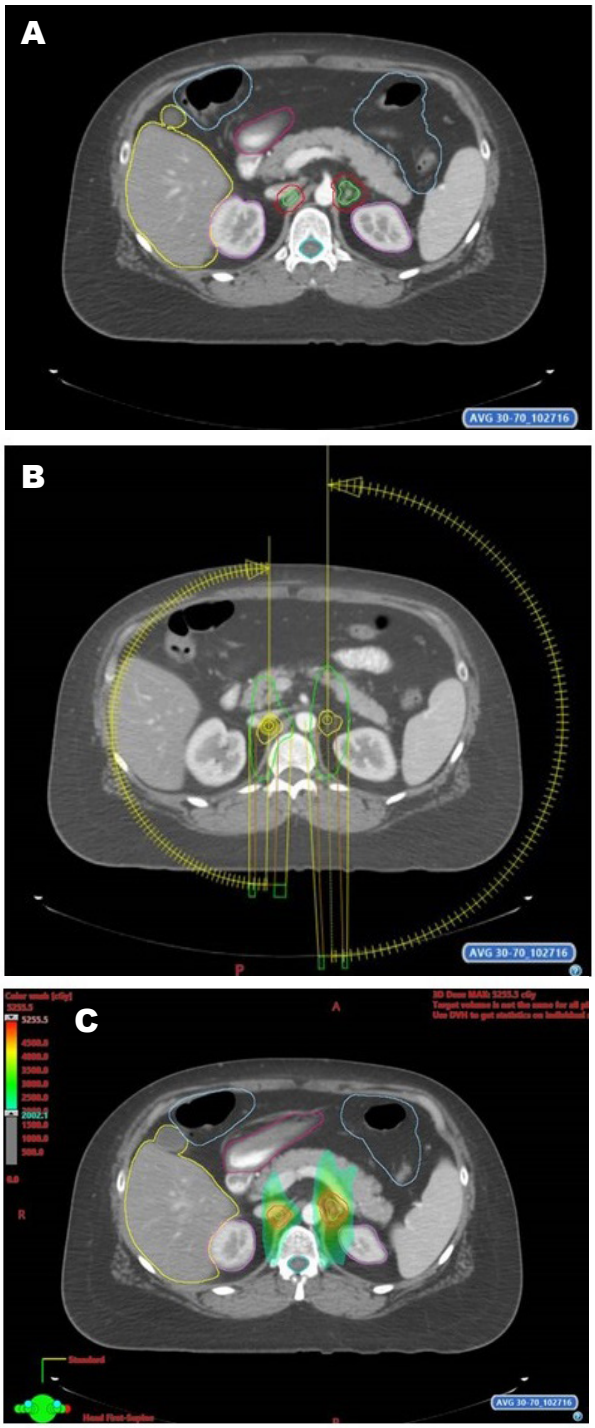


Fig. 1. Adrenal radiation planning depicted on computed tomography scan of the abdomen. *A*, Planned target volume (in red) for adrenal SBRT included a 0.5 cm safety margin around the radiologic abnormality (in green). Organs at risk for damage from radiation treatment (OARs) included bilateral kidneys (purple), spinal cord (turquoise), stomach (magenta), bowel (blue), and liver (yellow). *B*, Image shows orientation of stereotactic radiotherapy beam arcs. One clockwise and one counterclockwise partial-arc arrangement was utilized to treat each target. *C*, The adrenal beds were treated to 3,900 cGy in 3 fractions. Dose color-wash at approximately 50% level shows rapid dose fall-off and excellent sparing of OARs. *SBRT* = stereotactic body radiotherapy.

medical therapy is often indicated following RT, given the time required for the effects of RT to manifest.

Bilateral adrenalectomy is yet another second line therapy for CD. The advantage of BLA is that it offers prompt and reliable relief of hypercortisolemia. However, the disadvantage of BLA is that it mandates lifelong glucocorticoid and mineralocorticoid replacement (4). With a few exceptions, it is generally considered a last resort therapy after repeat TSS, RT, and medical therapy have failed. Nelson syndrome is a complication that occurs in 8 to 38% of patients who have undergone BLA and is caused by rapid enlargement of the remaining pituitary adenoma in response to elimination of negative feedback from endogenous adrenal production of cortisol (5,6). Symptoms related to tumor enlargement include headaches, visual field defects, and cranial nerve palsies. High circulating ACTH levels often lead to skin and mucosal hyperpigmentation due to stimulation of melanocytes. Furthermore, unrestricted ACTH stimulation can promote enlargement of ectopic or eutopic adrenal rests and lead to recurrent hypercortisolism (7,8). In terms of treating Nelson syndrome, stereotactic radiosurgery has demonstrated sustained efficacy in reducing tumor volume and decreasing ACTH levels (6). A number of medical therapies including sodium valproate, temozolomide, somatostatin analogs, dopamine agonists, and rosiglitazone have been studied in the treatment of Nelson syndrome, but have shown inconsistent results and require longer and larger studies to accurately assess their utility.

In CD cases where adrenalectomy is not desirable or possible, less invasive approaches to managing hyperfunctional adrenal tissue include radiofrequency ablation and adrenal radiation (9-11). Adrenal radiation is a well-established noninvasive treatment option for malignant metastases to the adrenal glands. Dosing is determined on the basis of tumor size and location, and ranges between 20 to 70 Gy in 1 to 15 fractions (12-14). The most common side effects are low grade gastrointestinal-related symptoms (nausea, diarrhea, abdominal pain) and fatigue. Low grade adrenal insufficiency has also been reported and is primarily seen in patients receiving radiation to bilateral adrenal glands (12,13). In their analysis of toxicities, Toesca et al (13) showed that SBRT had no short-term impact on renal function with a mean estimated decline in glomerular filtration rate of only 2.6 ± 8 mL/min/1.73 m^2 compared with baseline. SBRT to adrenal metastases is well tolerated, with nearly all reported side effects being grade 2 or less (12,13).

CONCLUSION

In patients with Nelson syndrome, any remnant or ectopic adrenal tissue can produce cortisol and lead to recurrence of CD. We have presented the first report of

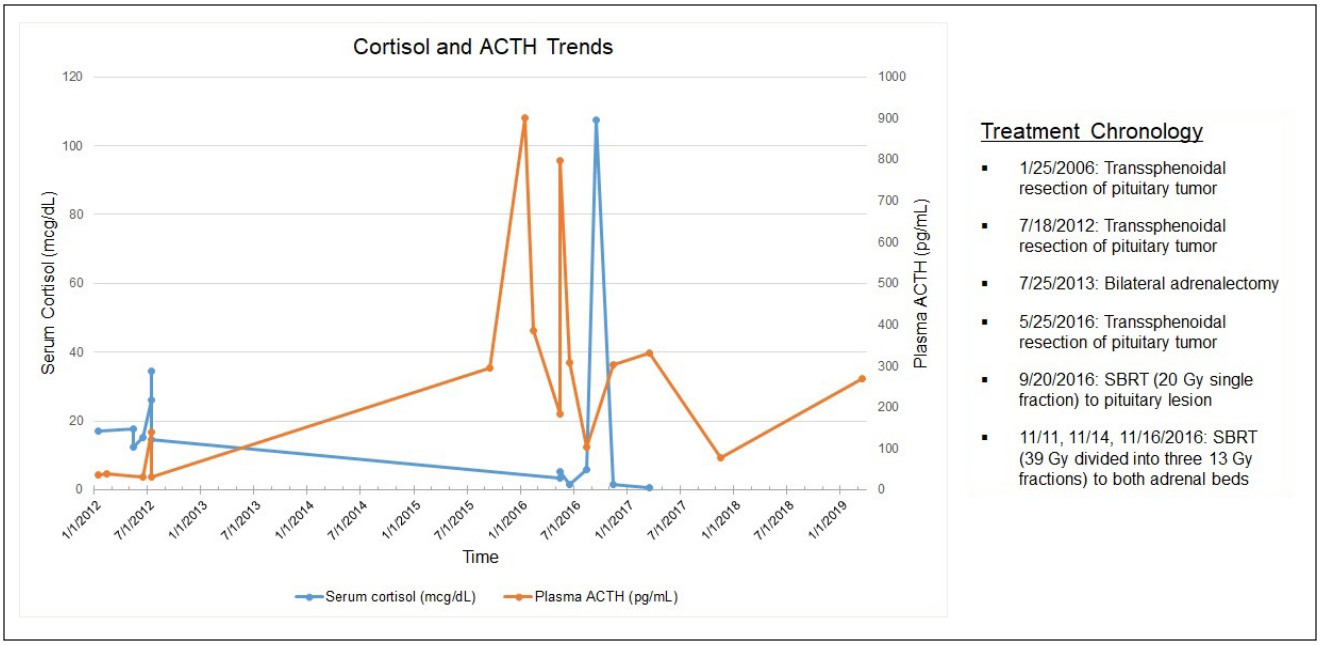


Fig. 2. Graph shows changes in serum cortisol and plasma ACTH levels over time. Treatment chronology included for reference. *ACTH* = adrenocorticotropic hormone; *SBRT* = stereotactic body radiotherapy.

adrenal ablation by radiotherapy in this setting, leading to a cure of CD with minimal adverse effects. Further studies are needed to evaluate the long-term efficacy of this treatment strategy.

DISCLOSURE

J.R.P. is a consultant for Intuitive Surgical and receives honoraria. T.B.V. has received research support from Corcept Therapeutics. The other authors have no multiplicities of interest to disclose.

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